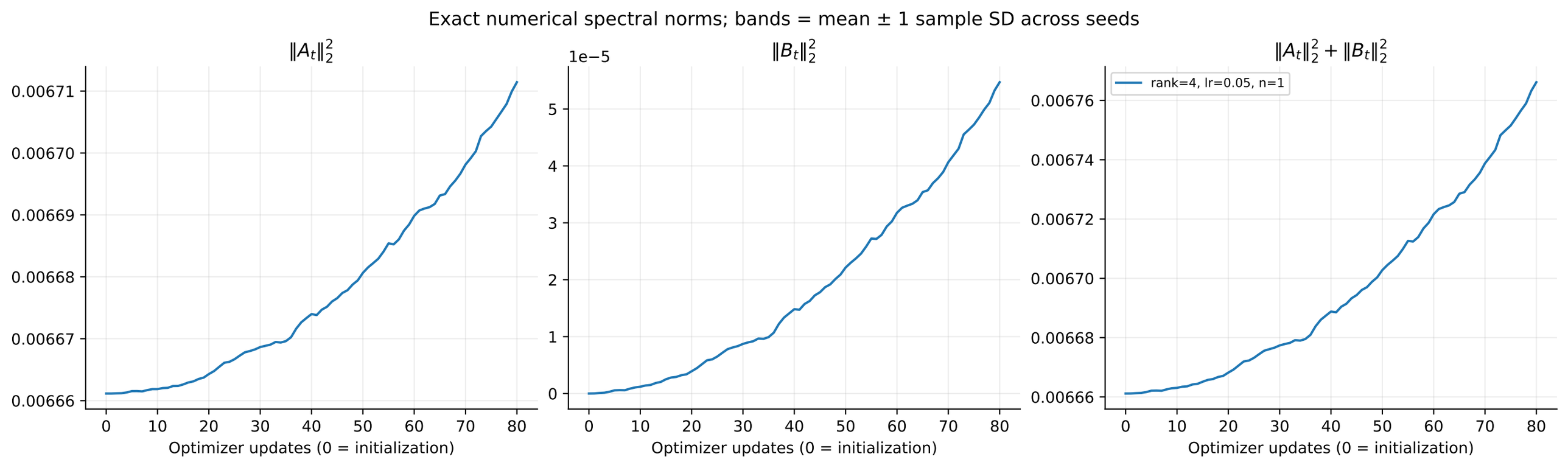
Exact numerical spectral norms; bands = mean ± 1 sample SD across seeds
$\|A_t\|_2^2$
$\|B_t\|_2^2$
$\|A_t\|_2^2 + \|B_t\|_2^2$
1e−5
rank=4, lr=0.05, n=1
0.00671
0.00670
0.00669
0.00668
0.00667
0.00666
0.00676
0.00674
0.00672
0.00670
0.00668
0.00666
5
4
3
2
1
0
0 10 20 30 40 50 60 70 80
Optimizer updates (0 = initialization)